Yale

Differential Privacy

CPSC 471 / 571: Trustworthy Deep Learning

Rex Ying

Content

- Introduction of Privacy
- Differential Privacy (DP)
- Differential Privacy in Deep Learning

Privacy – An Example

If the university asks you to take a survey if you have diabetes or not, **and the result will be released to the public**, would you participate in this survey?



Medical Research

Privacy – An Example

If the university asks you to take a survey if you have diabetes or not, **and the result will be released to the public**, would you participate in this survey?

Probably not



Medical Research

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If the university asks you to take a survey if you have diabetes or not, **and the result will be released to the public**, would you participate in this survey?

- **What if** they only ask for your age, location, birthday, and diseases you got in the past, no personally identifiable information?



Medical Research

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  - Not my name, bank account, or identity card. **So it's safe???**







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Not sure, why?

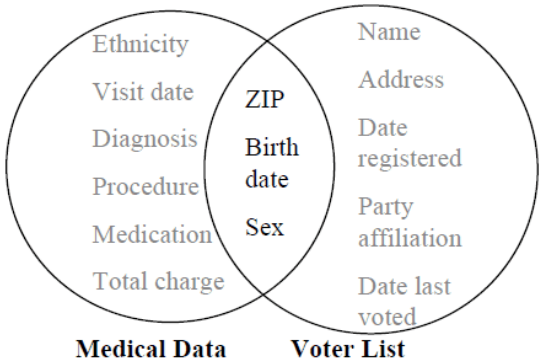


Medical Research

Linkage Attacks

- Connecting different datasets that seem to not relate to each other might reveal your identity.
- **The Commonwealth of Massachusetts Group Insurance Commission (GIC)** releases 135,000 records of patient encounters, each with 100 attributes.
  - Relevant attributes were removed, but **ZIP, birth date, and gender** were available.
  - It's usually considered as a "*safe*" practice.
- **Public voter registration record**
  - Contain, among others, name, address, **ZIP, birth date, and gender**

Linkage



87 % of the US population is uniquely identifiable by 5-digit ZIP, gender, and Date of Birth

The adversary can query the database using your ZIP code, birth date, and gender to see if you have diabetes or not.

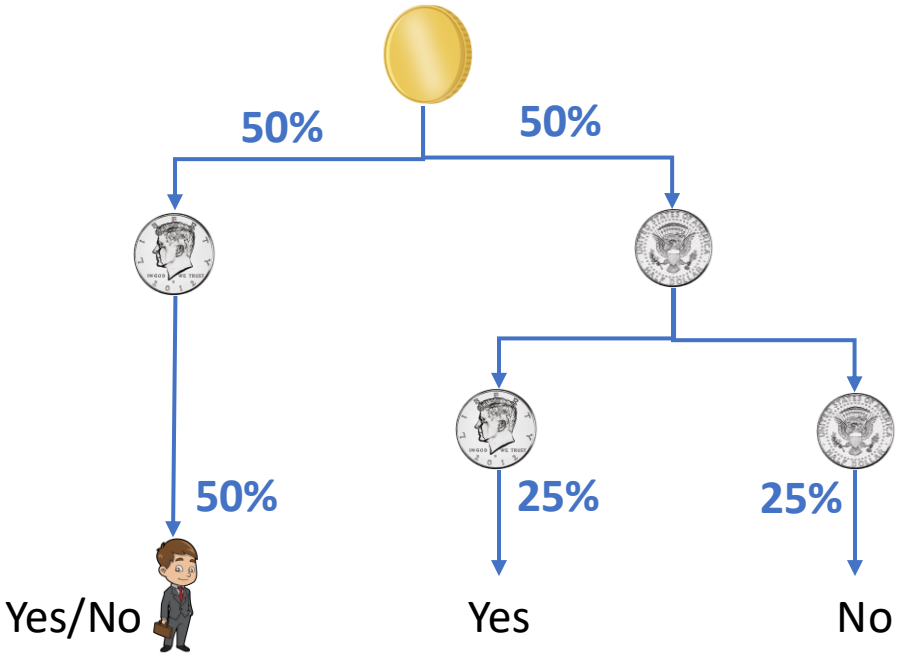
Privacy – An Example

- From the university's point of view

How can we provide a more rigorous guarantee to protect user's privacy?

- **Idea**: add random noise to users' answers
  - We **flip a coin** for each answer to determine if we collect the true user's answer.
  - If it's head, we record the user's true answer.
  - If it's tail, we record a random answer (50/50 the user could either have diabetes or not).

How does this strategy prevent adversaries from accessing private information?



Privacy – An Example

- From the university's point of view

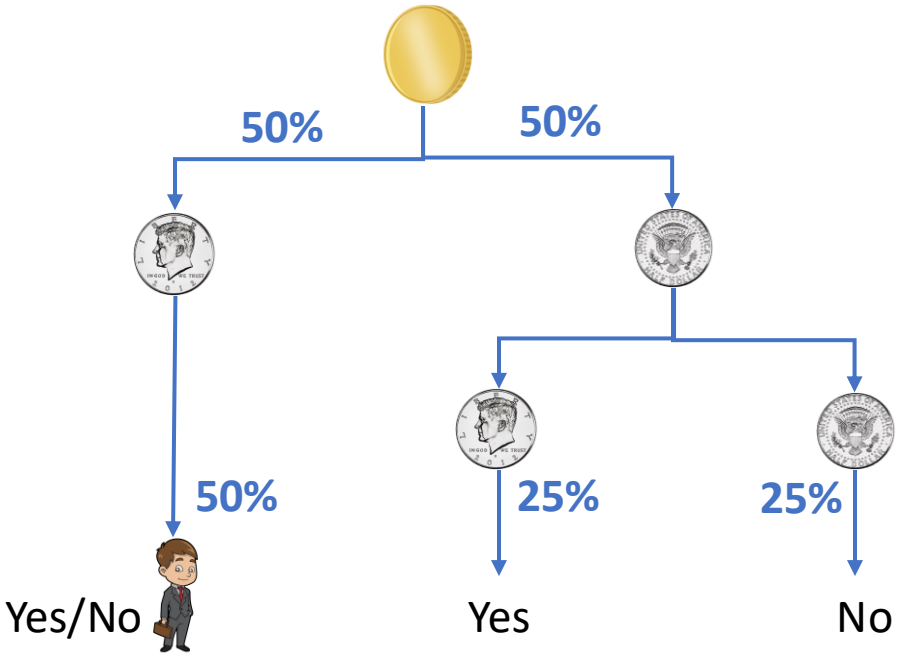
How can we provide a more rigorous guarantee to protect user's privacy?

- **Idea**: add random noise to users' answers
- **Do we still have meaningful statistics?**
  - **Yes, we can still know the true probability of people having diabetes**
  - $p_{T'} = \frac{1}{2} p_T + \frac{1}{4}$

$$\Rightarrow p_T = 2p_{T'} - \frac{1}{2}$$

(true) probability of a person having diabetes.

(noisy) probability of a person having diabetes according to the survey.

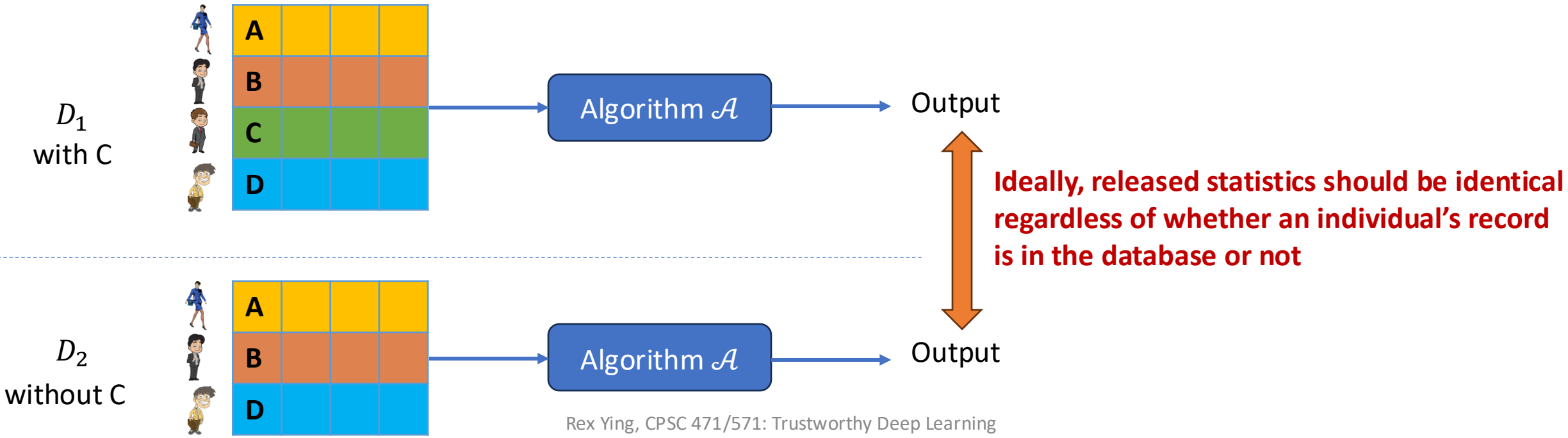


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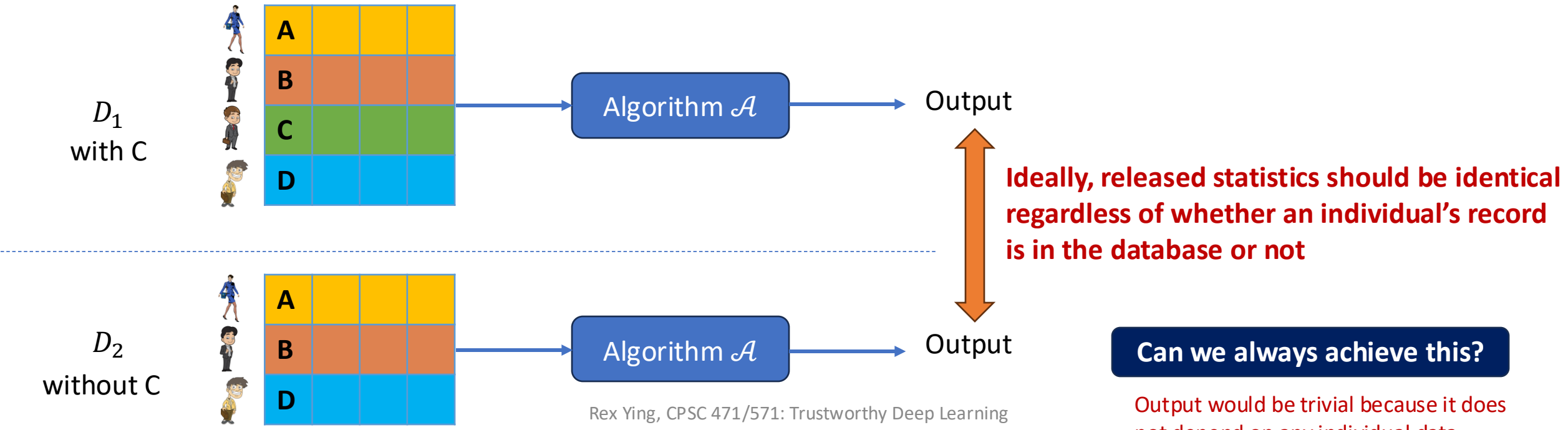
Differential Privacy

- **Cynthia Dwork (2006)** proposes a formal definition of *individual* privacy:
  - **Intuition:** Any information-related risk to a person should not change due to that person's information being included or not in the analysis.



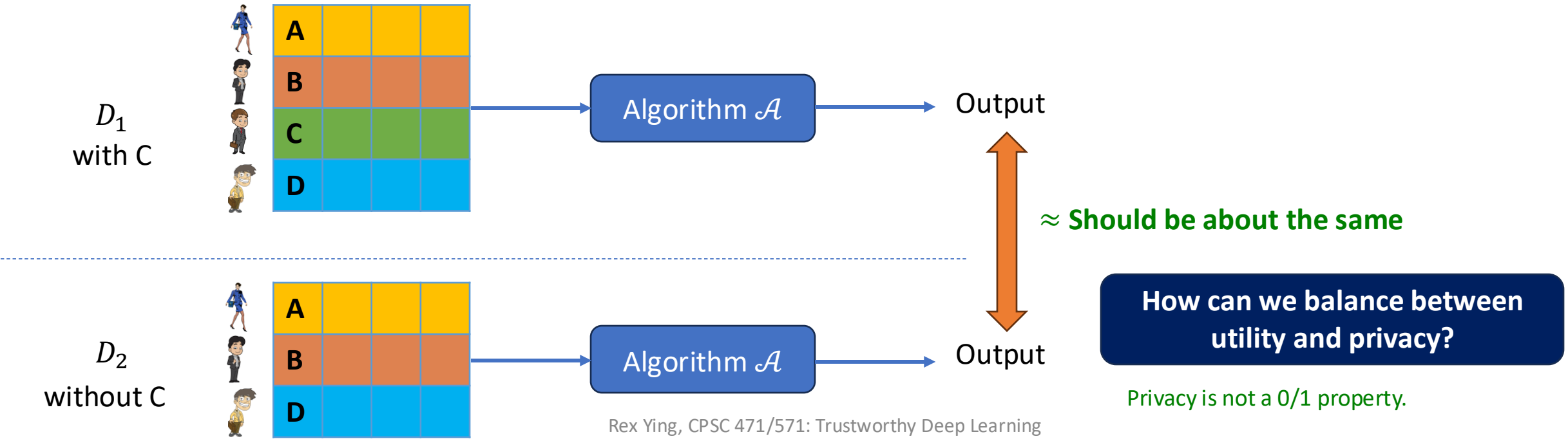
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Differential Privacy

- **Cynthia Dwork (2006)** proposes a formal definition of *individual* privacy:
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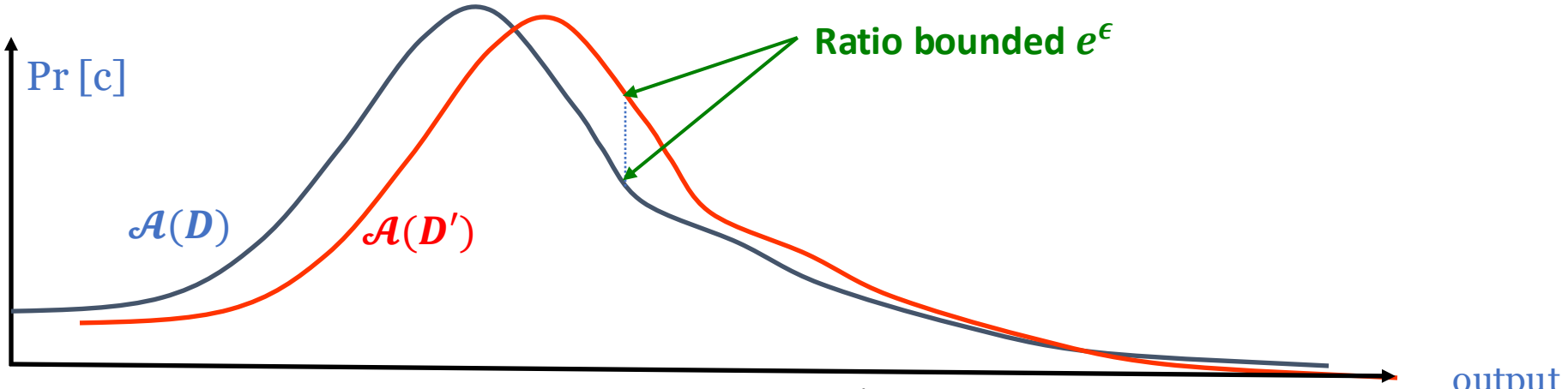
ϵ –Differential Privacy

Definition [ϵ – Differential Privacy] For $\varepsilon \geq 0$, an algorithm $\mathcal{A}$ is **ε-differentially private** if and only if for any pair of neighboring datasets **D and D' that differ in only one element** and any $C \subseteq \text{range}(\mathcal{A})$:

$$\Pr[\mathcal{A}(D) \in C] \leq e^{\varepsilon} \Pr[\mathcal{A}(D') \in C], \qquad \forall C$$

where $\Pr[\mathcal{A}(D) \in C]$ denotes the probability that the algorithm $\mathcal{A}$ outputs $c \in C$.

- **Derived differential privacy loss**: $\ln \frac{\Pr[\mathcal{A}(D) \in C]}{\Pr[\mathcal{A}(D') \in C]} \leq \varepsilon.$



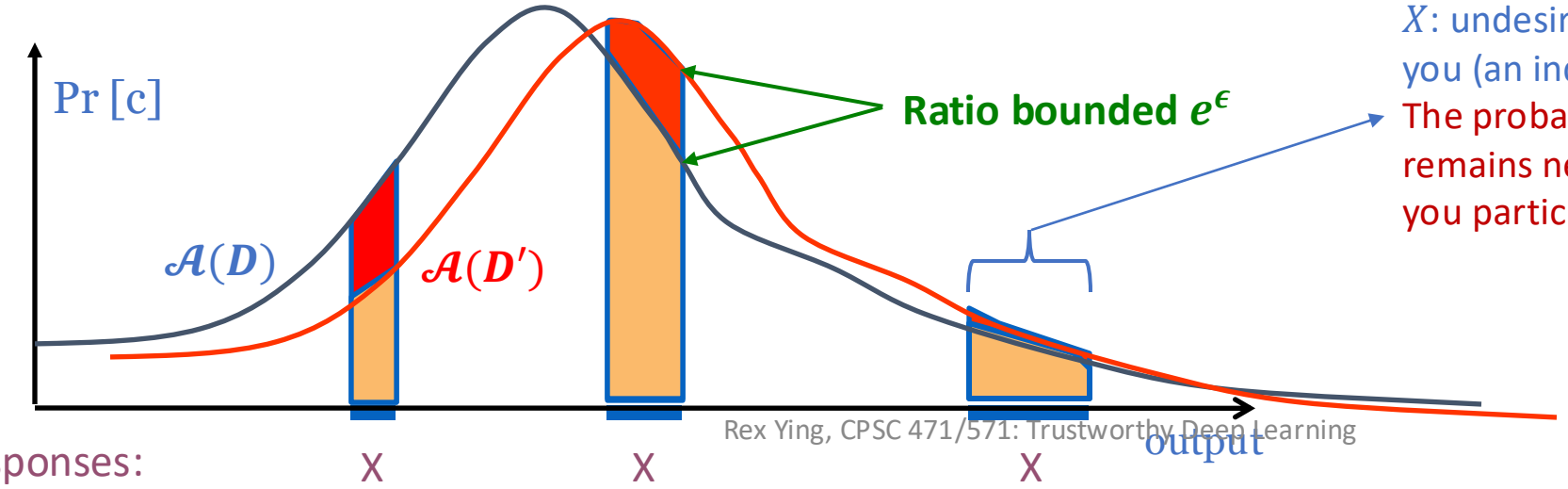
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where $\Pr[\mathcal{A}(D) \in C]$ denotes the probability that the algorithm $\mathcal{A}$ outputs $c \in C$.

- **Implication on differential privacy**: Anything an adversary can do to you, it could do without your data.



ϵ −Differential Privacy

Definition [ϵ −Differential Privacy]

$$\Pr[\mathcal{A}(\boldsymbol{D}) \in \boldsymbol{C}] \leq \boldsymbol{e^{\varepsilon}} \Pr[\mathcal{A}(\boldsymbol{D'}) \in \boldsymbol{C}], \qquad \forall \subseteq \text{range}(\mathcal{A})$$

Is there any drawback associated with this definition?

ϵ −Differential Privacy

Definition [ϵ −Differential Privacy]

$$0 = \Pr[\mathcal{A}(D) \in C] \leq e^{\varepsilon} \Pr[\mathcal{A}(D') \in C], \quad \forall C \subseteq \text{range}(\mathcal{A})$$

$0 =$: $=0$

Is there any drawback associated with this definition??

- **If** an ϵ −differentially private algorithm $\mathcal{A}$ has probability zero at any output c for the dataset $D', i.e., \Pr[\mathcal{A}(D') = c] = 0,$
- **then** the algorithm $\mathcal{A}$ has probability zero at c for every other dataset D.

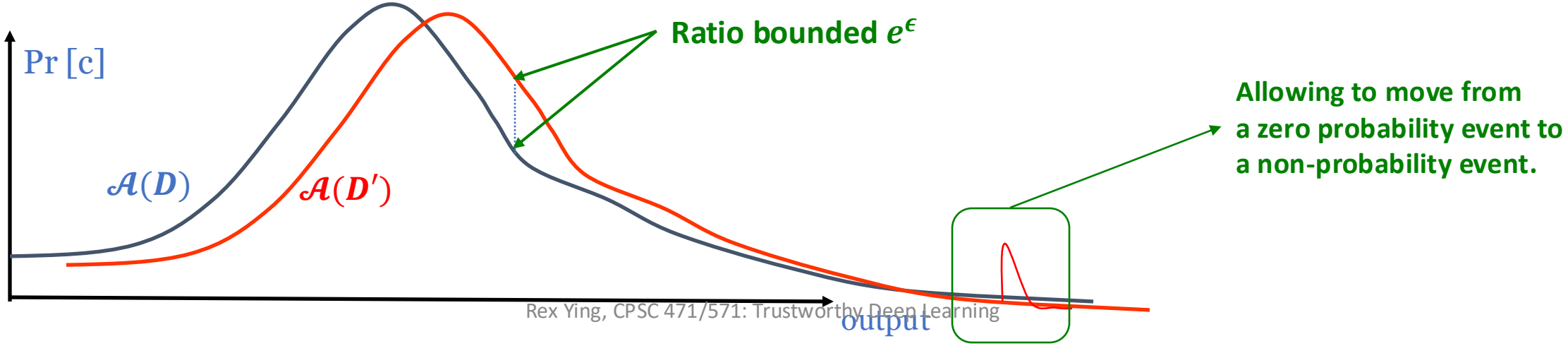
→ Too strong?

(ϵ, δ) −Differential Privacy

Definition [$(\epsilon, \boldsymbol{\delta})$ −Differential Privacy]

$$\Pr[\mathcal{A}(\boldsymbol{D}) \in \boldsymbol{C}] \leq \boldsymbol{e^{\varepsilon}} \Pr[\mathcal{A}(\boldsymbol{D'}) \in \boldsymbol{C}] + \boldsymbol{\delta}, \qquad \forall C \subseteq \text{range}(\mathcal{A})$$

- **Solution:** We can relax this constraint by allowing an additive difference $\boldsymbol{\delta}$ between two subsets of data, along with the multiplicative factor ($\boldsymbol{e^{\epsilon}}$).
- **Derived differential privacy loss:** $\ln \frac{\Pr[\mathcal{A}(D) \in C] - \delta}{\Pr[\mathcal{A}(D') \in C]} \leq \varepsilon.$



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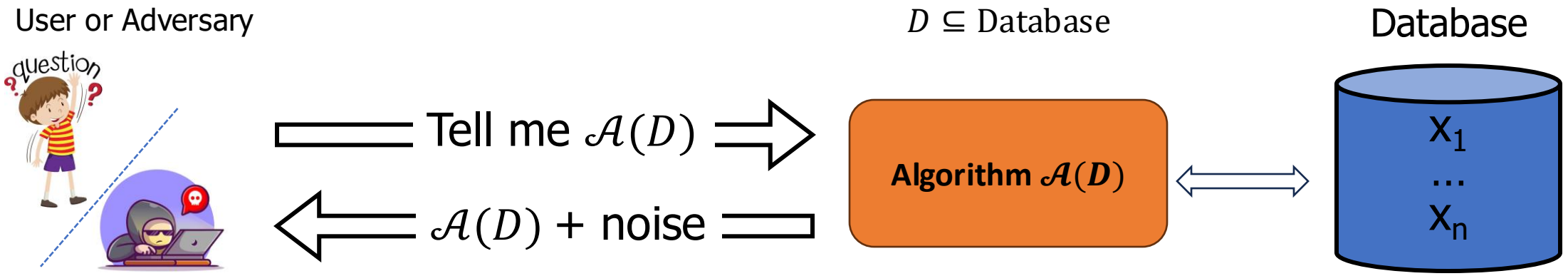
Why it intuitively makes sense?

- **Remarks**: Differential privacy is a **statistical property** of **algorithmic mechanisms**.
  - DP remains robust against auxiliary information (*e.g.,* voting database in linkage attack example).
  - The guarantee of DP is independent of the computational capabilities of a potential adversary.

How can we inject differential privacy into an algorithm?

Design Differentially Private Algorithms

- **Idea (again)**: add random noise to algorithm outputs.



- **How much noise is needed?**
  - **Intuition:**
    - Add ***more noise*** if $\mathcal{A}$ is ***more sensitive*** to individual data (may contain more private information).
    - Add ***less noise*** if $\mathcal{A}$ is ***less sensitive*** to individual data.

What explainability method does it remind you of?

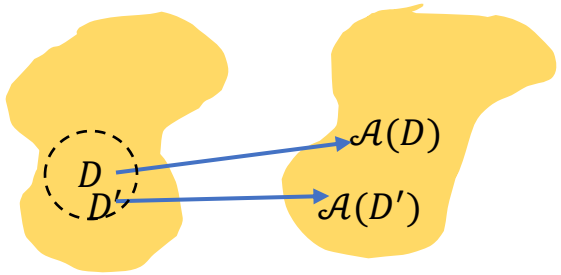
Laplace Mechanism → ϵ-DP

Definition (Global sensitivity): **Global sensitivity** of a function $\mathcal{A}$, denoted Δ, is

$$\Delta = \sup_{D,D': ||D \setminus D'|| \leq 1} |\mathcal{A}(D) - \mathcal{A}(D')|$$

D, D' differ in only one element.

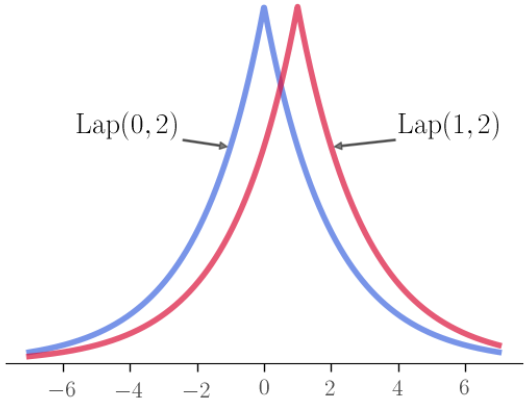
Theorem (Laplace Mechanism): Given a function $\mathcal{A}$, a dataset D and a fixed $\varepsilon \geq 0$, the randomizing algorithm $\mathcal{A}_{DP}(D) = \mathcal{A}(D) + Z$ satisfies **ε-DP**, where Z is a random variable from a **Laplace distribution**, i.e., $Z \sim Lap(0, \frac{\Delta}{\varepsilon})$.



$$Z \sim Lap(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$

$$\Rightarrow \mathbb{E}(|Z|) = b \quad \& \quad \Pr(|Z| \geq t \cdot b) = e^{-t}$$

Laplace mechanism offering **0.5-differential privacy** for a function with **sensitivity (b) 1**.



Source: wikipedia

Gaussian Mechanism → (ϵ, δ)-DP

Definition (ℓ_2-sensitivity): **ℓ_2-sensitivity** of a function $\mathcal{A}$, denoted Δ_2, is

$$\Delta_2 = \sup_{D,D': ||D - D'|| \le 1} ||\mathcal{A}(D) - \mathcal{A}(D')||_2$$

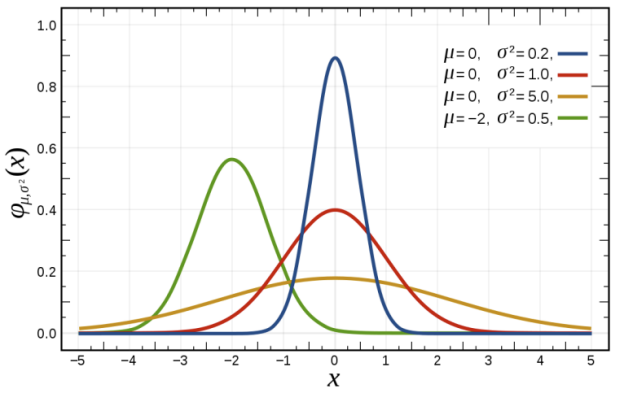
D, D' differ in only one element.

Theorem (Gaussian Mechanism): Given a function $\mathcal{A}$, a dataset D and a fixed $\varepsilon, \boldsymbol{\delta} \geq 0$, the randomizing algorithm $\mathcal{A}_{DP}(D) = \mathcal{A}(D) + Z$ satisfies **(ε, δ)-DP**, where Z is a random variable from a **Gaussian distribution**, i.e., $Z \sim \mathcal{N}(0, \frac{2\Delta_2^2 \ln(\frac{1.25}{\delta})}{\epsilon^2})$.

- Exponential mechanism is also commonly used in literature but will not be discussed in this lecture.

$$Z \sim \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

$$\Rightarrow \mathbb{E}(Z) = 0 \quad \& \quad \Pr(|Z| \geq t) \leq 2e^{-t^2/(2\sigma^2)}$$



See proof

Source: wikipedia

Properties of Differential Privacy

Prop 1 (Sequential Composition): Let $\mathcal{A}_1 + \mathcal{A}_2$ is an (ϵ_1, δ_1)-DP algorithm $\mathcal{A}_1$ followed by an (ϵ_2, δ_2)-DP algorithm $\mathcal{A}_2$. Then $\mathcal{A}_1 + \mathcal{A}_2$ is $(\epsilon_1 + \epsilon_2, \delta_1 + \delta_2)$-DP.

- **Implication**: Differential privacy loss scales with the number of queries, *i.e.*, if we run an (ϵ, δ)-DP algorithm k times, the accumulated DP loss is $(k\epsilon, k\delta)$-DP.

- If we *do not allow* any slack in δ, this bound cannot be tightened.

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- If we *do not allow* any slack in δ, this bound cannot be tightened.
  If we *allow* a slack in the additive factor, we can get a higher privacy.

Prop 1.1 (Advanced Composition): For any $\epsilon > 0, \delta \in [0, 1], \tilde{\boldsymbol{\delta}} \in (0, 1]$, an (ϵ, δ)-DP algorithm $\mathcal{A}$ satisfies $\left(\tilde{\boldsymbol{\epsilon}}, k\delta + \tilde{\boldsymbol{\delta}}\right)$-DP under k-fold composition, where

$$\tilde{\epsilon} = k\epsilon(e^{\epsilon} - 1) + \epsilon\sqrt{2k\log(1/\tilde{\delta})}$$

See proof

$$\approx O(k\epsilon^2 + \sqrt{k\epsilon^2}) \ll O(k\epsilon) \quad \text{if } \epsilon \ll 1$$

Properties of Differential Privacy

Prop 2 (Parallel Composition): If $\mathcal{A}$ is (ϵ,δ)-differentially private on datasets X. Then an algorithm releasing $\mathcal{A}(D_1), \mathcal{A}(D_2), \dots, \mathcal{A}(D_k)$ is (ϵ,δ)-differentially private if $D_1, D_2, \dots, D_k$ are disjoint sets in X

- **Implication**: splitting your dataset into disjoint chunks and running a differentially private mechanism on each chunk separately does ***not*** increase your DP loss.

Prop 3 (Post-processing): If $\mathcal{A}$ is (ϵ, δ)-differentially private, then for any (deterministic or randomized) function g, $g(\mathcal{A})$ satisfies (ϵ, δ)-differential privacy

- **Implication**: DP is independent of the adversary's power. In other words, it is impossible to reverse the privacy protection provided by differential privacy by post-processing the output in some way.

Further Readings

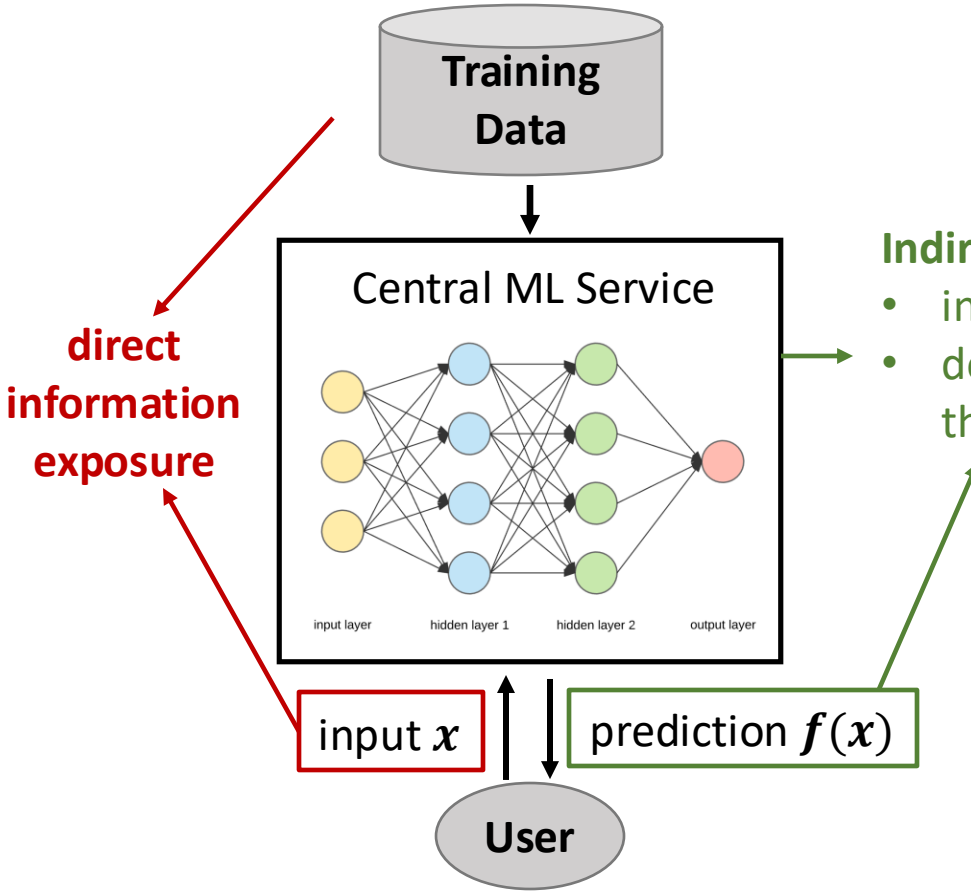
- **Most of the bounds** on the privacy cost we have seen so far are *upper* bound and they *sometimes* are very loose bounds.

- There are some variants of DP to enable tighter bounds on the privacy cost, especially for iterative algorithms, e.g., Rényi differential privacy and Zero-concentrated differential privacy:

Content

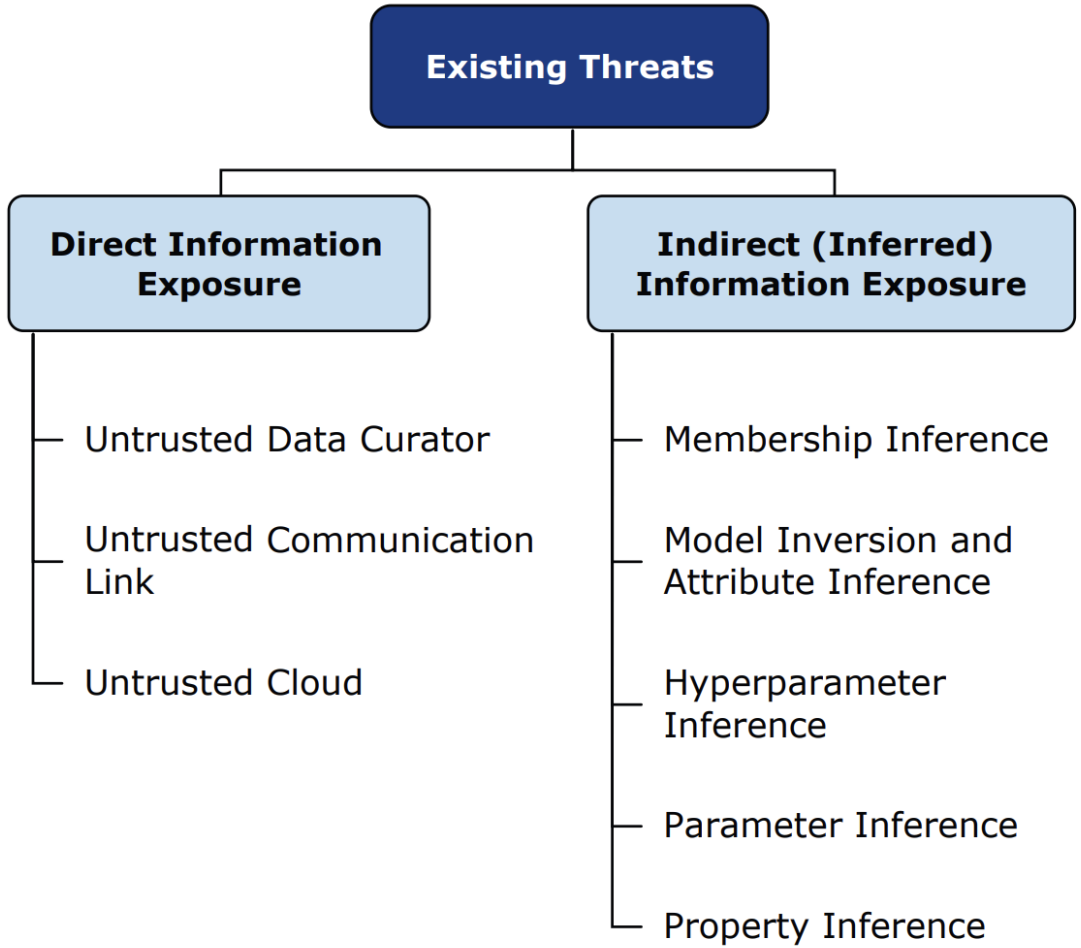
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ML Model Are Not Safe



Indirect leakage:

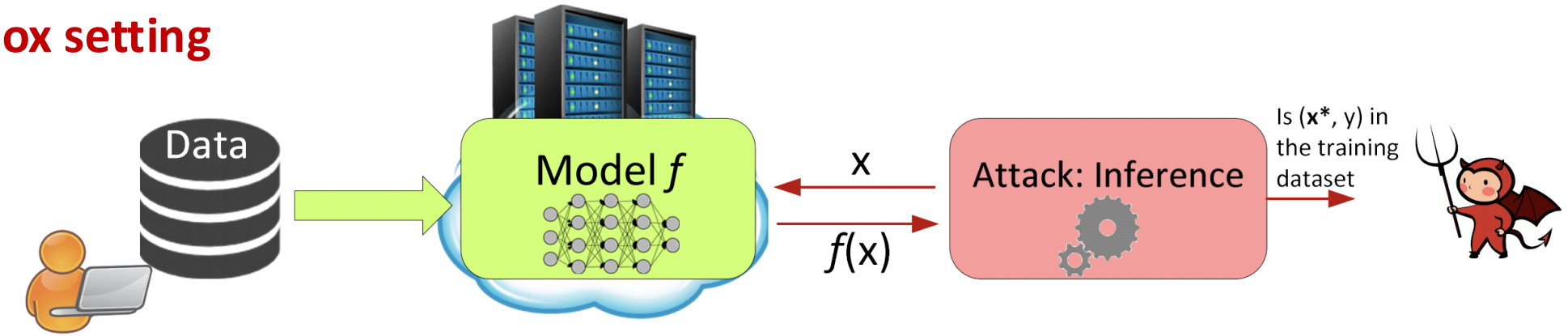
- infer the information
- do not have access to the actual data



Membership Inference Attack (1)

- **Goal:** speculate whether a given data instance is part of the training dataset of a target model
- **Main idea:** train several **shadow models** that mimic the behavior of the target model. Then **use these shadow models to train a binary classifier** to **infer the membership** of the target model.

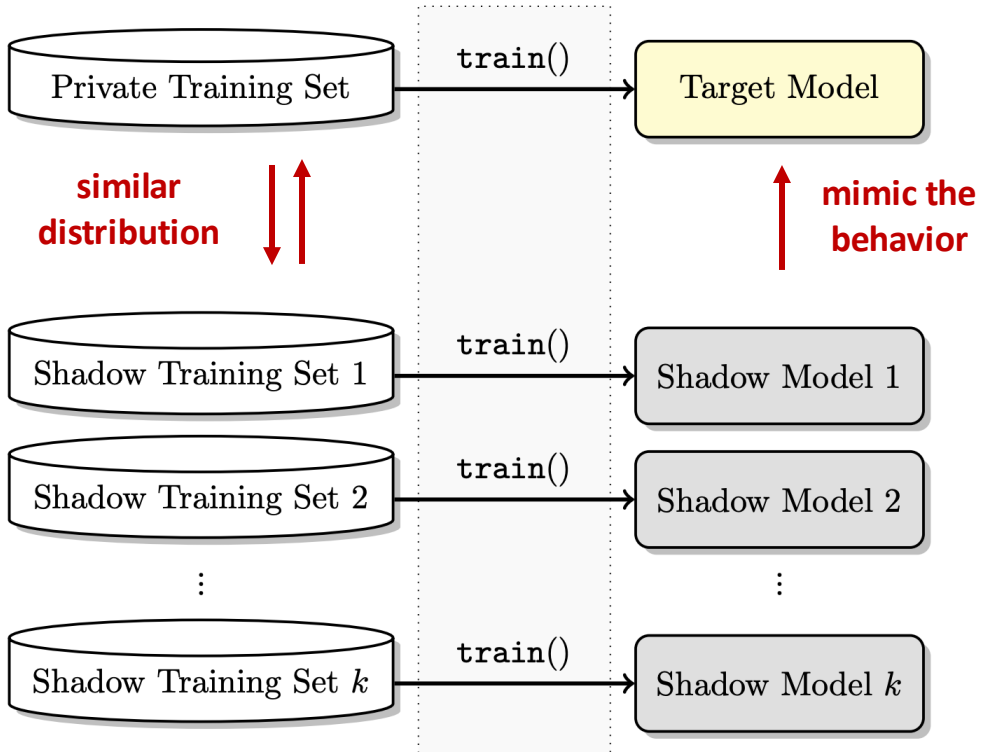
Blackbox setting



Liu, Bo, et al. "When machine learning meets privacy: A survey and outlook."

Membership Inference Attack (2)

- **Shadow Training Set:** has a **similar distribution** as the Private Training Set
- How to generate shadow training sets?
  - **Model-based synthesis:** records the data that are classified by the target model with high confidence
  - **Statistics-based synthesis:** the attacker knows some statistical information about the dataset
  - **Noisy real data:** the attacker has access to the noisy version of the real data

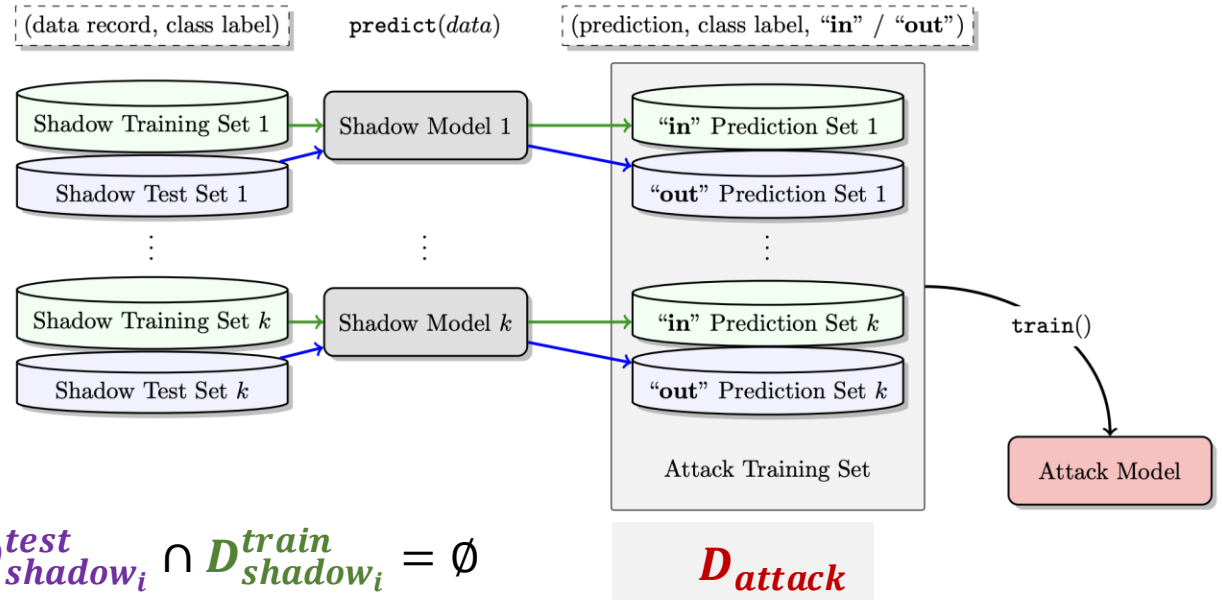


Shokri, Reza, et al. "Membership inference attacks against machine learning models."

Membership Inference Attack (3)

- Use the **output probability vectors** from the shadow models to train the attack model (**binary classifier**)

- Shadow test set $D^{test}_{shadow_i}$ is **disjoint** from the shadow training set $D^{train}_{shadow_i}$
- D_{attack}: training set for the **Attack Model**
- $f_{shadow_i}(\,)$: the i-th shadow model
  - For $(x, y) \in D^{train}_{shadow_i}$, add $(y, f_{shadow_i}(x), in)$ to D_{attack}
  - For $(x, y) \in D^{test}_{shadow_i}$, add $(y, f_{shadow_i}(x), out)$ to D_{attack}



$D^{test}_{shadow_i} \cap D^{train}_{shadow_i} = \emptyset$

D_{attack}

Attack Model: predict whether individual instances are in the private training set of the target model

Model Inversion and Attribute Inference

- **Goal: infer sensitive attributes** of a given data instance from the non-sensitive attributes of the instance and the target model
- **Main idea:** apply **gradient descent** on the input to **maximize the logit of the target label**
  - Example: given a person's name and access to a facial recognition system, the attacker recovers an averaged image that makes the person recognizable



Recovered image using **model inversion attack**



True image in the training set

Algorithm 1 Inversion attack for facial recognition models.

1: **function** MI-FACE($label, \alpha, \beta, \gamma, \lambda$)
2: $c(\mathbf{x}) \stackrel{\text{def}}{=} 1 - \tilde{f}_{label}(\mathbf{x}) + \text{AUXTERM}(\mathbf{x})$
3: $\mathbf{x}_0 \leftarrow \mathbf{0}$
4: **for** $i \leftarrow 1 \ldots \alpha$ **do**
5: $\mathbf{x}_i \leftarrow \text{PROCESS}(\mathbf{x}_{i-1} - \lambda \cdot \nabla c(\mathbf{x}_{i-1}))$
6: **if** $c(\mathbf{x}_i) \geq \max(c(\mathbf{x}_{i-1}), \ldots, c(\mathbf{x}_{i-\beta}))$ **then**
7: **break**
8: **if** $c(\mathbf{x}_i) \leq \gamma$ **then**
9: **break**
10: **return** $[\arg\min_{\mathbf{x}_i}(c(\mathbf{x}_i)), \min_{\mathbf{x}_i}(c(\mathbf{x}_i))]$

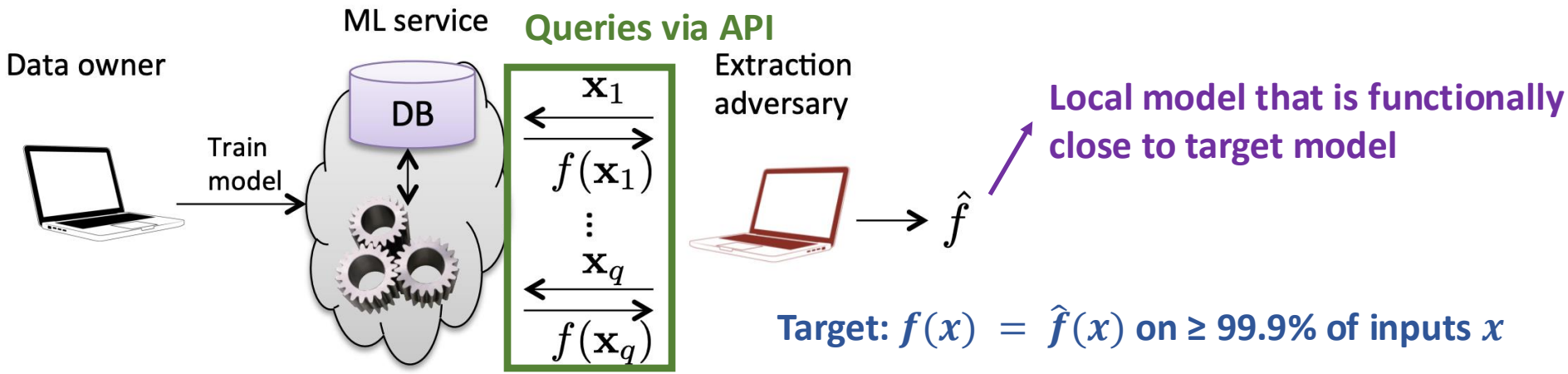
Optimize x, such that $\tilde{f}_{label}(x) \rightarrow 1$

Limitations?

Fredrikson, et al. "Model inversion attacks that exploit confidence information and basic countermeasures."

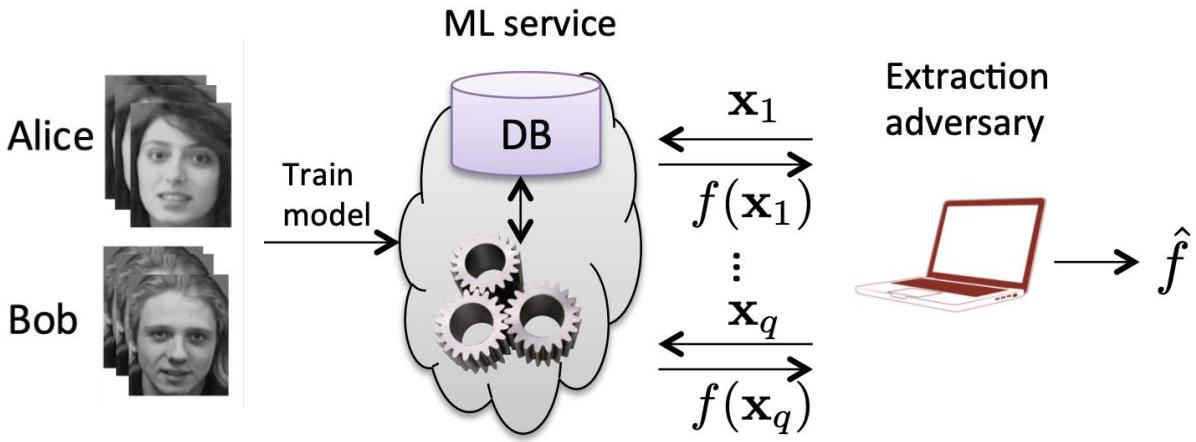
Model Extraction Attack

- **Model Extraction** reconstructs **an approximation model $\hat{f}(x)$** of the target model $f(x)$, including:
  - recover the **model parameters** via black-box access to the target model
  - find the **hyperparameters** used in the model training, e.g., number of layers, activation function, regularization coefficients, etc.



Example: Extraction of Logistic Regression

- Task: binary classification with logistic regression
- **Model distillation** (recall defensive distillation in adversarial defense) in its simplest form



$f(x) > 0.5 \Rightarrow$ Alice
$f(x) \leq 0.5 \Rightarrow$ Bob

Assume x **has** n **features,** then model has $n+1$ **unknown parameters** (n for w and 1 for b)

$$f(x) = \frac{1}{1+e^{-(w*x+b)}}$$

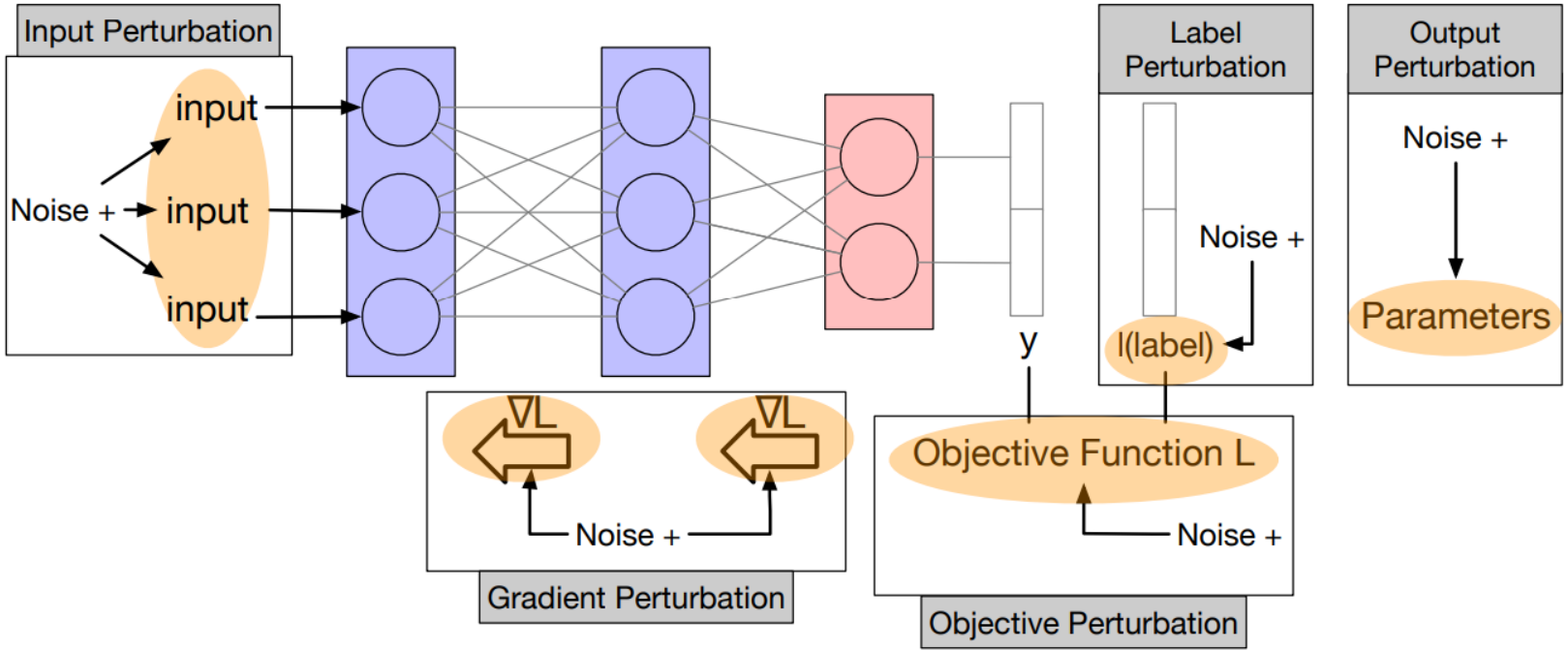
$$\ln\left(\frac{f(x)}{1-f(x)}\right) = w * x + b$$

Linear equation with $n+1$ unknows

Query $n+1$ predictions with random samples $\Rightarrow$ **solve a linear system of $n+1$ equations**

Differential Privacy in Machine Learning

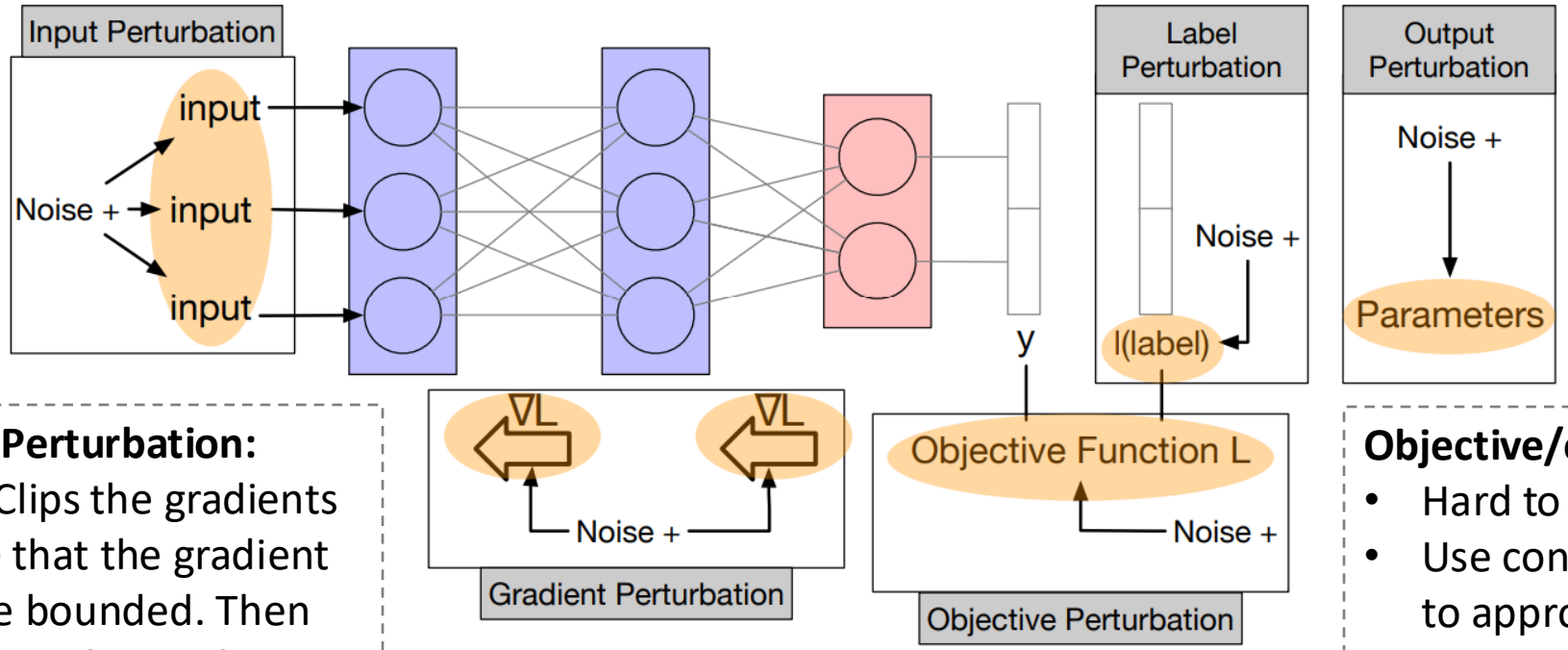
- Differential Privacy is achieved by applying noise mechanisms to
  - **Input data**
  - **Training stage**
    - Objective perturbation
    - Gradient perturbation
  - **Inference stage**
    - Label perturbation
    - Output perturbation



Pipeline of Differential Pravicy

Input perturbation: a type of Data Anonymization

Label Perturbation:
Used in [teacher ensemble method (PATE)]
Add noise to the labels of the teacher models and aggregate labels to train a student model



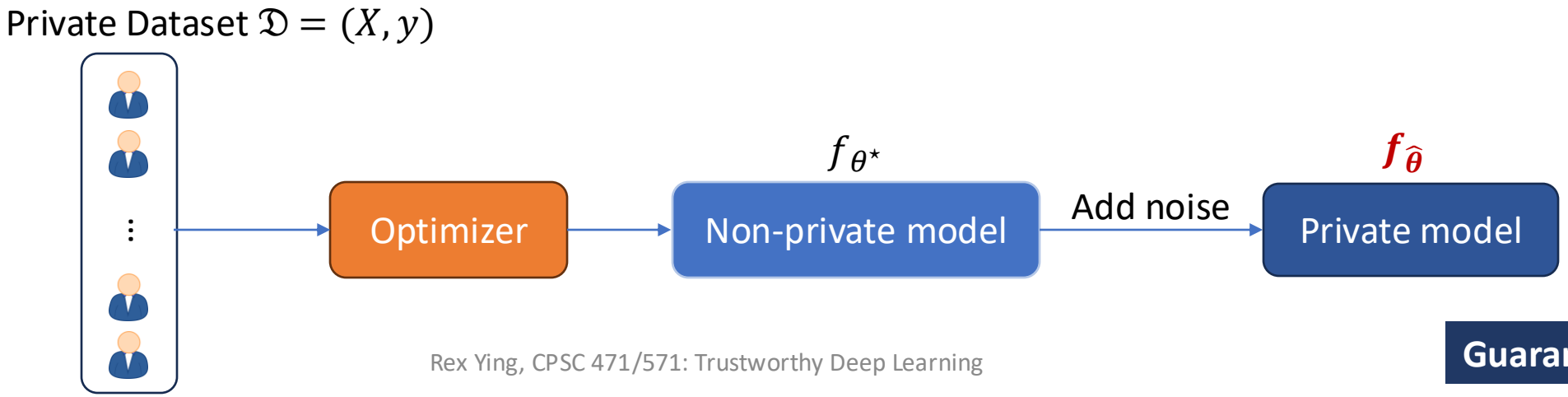
Gradient Perturbation:
[DP SGD]: Clips the gradients to ensure that the gradient norms are bounded. Then **add noise to the gradients**

Objective/output perturbation:
- Hard to calculate the sensitivity
- Use convex polynomial functions to approximate the target model

[reference]

Differential Privacy at Training Stage: DP-ERM

- **How to train** a differentially private model to mitigate membership inference attacks and model inversion attacks?
- **Naïve approach:** Differentially Private Empirical Risk Minimization (DP-ERM)
  1. Train an ML model to get optimal parameters $\theta^\star = \text{argmin}_\theta \mathcal{L}(f, \theta, \mathfrak{D})$.
  2. Add noise to the optimal parameters to perturb the model's output $\widehat{\boldsymbol{\theta}} = \theta^\star + Y$, where Y is a Gaussian noise $Y \sim \mathcal{N}(0, \sigma^2)$.



Guarantee?

Differential Privacy at Training Stage: DP-ERM

- **Privacy guarantee for DP-ERM**: applying Gaussian Mechanism

Gaussian Mechanism

If

- the algorithm $\mathcal{A}$ is Δ_2-sensitive (Δ_2 can be bounded),
- the model $\mathcal{A}_{DP}(D) = \mathcal{A}(D) + Z$ where $Z \sim \mathcal{N}\left(0, \frac{2\Delta_2^2 \ln\left(\frac{1.25}{\delta}\right)}{\epsilon^2}\right)$.

Then

- Algorithm $\mathcal{A}_{DP}(D)$satisfies (ε, δ)-DP.

DP-ERM

If

- the loss function $\mathcal{L}(f, \theta, \mathfrak{D})$ is 1-Lipschitz,
- the model $f_{DP,\hat{\theta}}$, where $\hat{\theta} = \theta^\star + Y$ where $Y \sim \mathcal{N}\left(0, \frac{8\ln\left(\frac{1.25}{\delta}\right)}{n^2\epsilon^2}\right)$ where $n = |\mathfrak{D}|$,

Then

- Training algorithm $\mathcal{A}_{f,\mathcal{L}}$ satisfies (ε, δ)-DP.

See proof

Definition (Lipschitz):

A function $\mathcal{L}$ is L-**Lipschitz w.r.t.** a norm (e.g. L2 norm in this case) $||\cdot||$ if and only if:

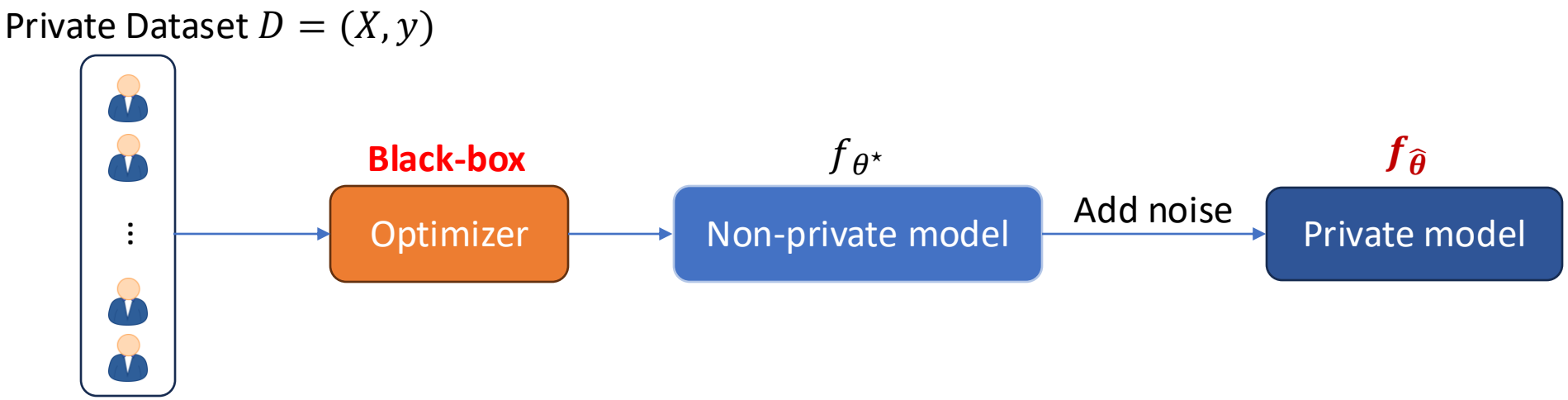
$$|\mathcal{L}(\theta) - \mathcal{L}(\theta')| \leq L||\theta - \theta'|| \qquad \forall\, \theta, \theta' \in \Theta$$

If we use L2 norm, the above is equivalent to $||\nabla\mathcal{L}(\theta)||_2 \leq L \quad \forall\, \theta \in \Theta$.

Limitations?

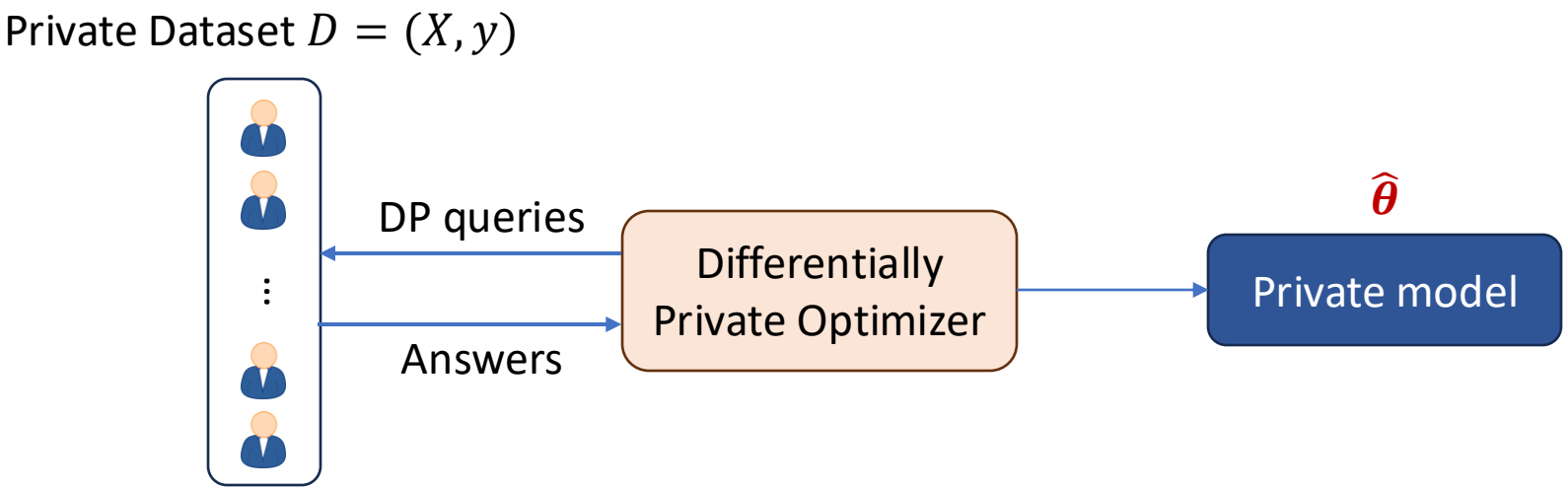
Differential Privacy at Training Stage: DP-ERM

- **Limitations**
  - It requires restrictive assumptions on the loss function.
  - The sensitivity is likely to be pessimistic (loose bound) as it treats ERM as a black box



Differential Privacy at Training Stage: DP-SGD

- **Differentially Private Stochastic Gradient Descent**
  - The idea is to perturb only the quantities needed for the optimizer
    → Perturb the gradient instead of perturbing the optimal model parameters.
  - Applying to Minibatch Stochastic Gradient Descent, which is already a randomized algorithm.



Amplification by Subsampling

- The following theorem suggests a privacy amplification via subsampling

Theorem (Amplification by subsampling): Let $\mathfrak{D}$ be a data domain and $\mathcal{S}: \mathfrak{D}^n \to \mathfrak{D}^m$ be a procedure returning a random mini-batch of data points size m sampled uniformly without replacement from a dataset $\mathfrak{D}$. Let $\mathcal{A}$ be a (ε, δ)-DP algorithm. Then $\mathcal{A} \circ \mathcal{S}$ satisfies $\left(\varepsilon', \frac{m}{n}\delta\right)$-DP, where $\varepsilon' = \underline{ln\left(1 + \frac{m}{n}(e^{\varepsilon} - 1)\right)} \leq 2\frac{m}{n}\varepsilon$ if $\varepsilon \leq 1$ See proof

- **Implication:** A differentially private mechanism runs on a random subsample of a population provides **higher** privacy guarantees than when run on the entire population.
- **More data samples** (larger n), higher privacy
- **Less batch size** (smaller m), higher privacy

Subsampling is good for privacy ☺

Differential Privacy at Training Stage: DP-SGD

- **Perturbing the gradient update of SGD to provide a privacy guarantee**

Vanilla SGD

- Initialize parameters $\theta^{(0)} \in \Theta$
- For $t = 0, \dots, T-1$:
  - Choose a sample $x_i \in \mathfrak{D}$ uniformly
  - Update gradient
    $\theta^{(t+1)} = \text{Proj}\left(\theta^{(t)} - \gamma \nabla \mathcal{L}(f, \theta^{(t)}, x_i)\right)$
- Return $\theta^{(T)}$

DP-SGD

- Initialize parameters $\theta^{(0)} \in \Theta$ **(independent of D)**
- **For $t = 0, \dots, T-1$:** → Advanced composition
  - **Choose a sample $x_i \in \mathfrak{D}$ uniformly** → Amplification by subsampling with batch size = 1
  - **Random noise $\eta^{(t)} \sim \mathcal{N}(0, \sigma^2)$ where**

$$\sigma^2 = \frac{16^2 L^2 T \ln\left(\frac{2}{\delta}\right) \ln\left(\frac{2.5T}{\delta n}\right)}{n^2 \epsilon^2}$$ → Gaussian mechanism

  - Update gradient
    $\theta^{(t+1)} = \text{Proj}\left(\theta^{(t)} - \gamma\left(\nabla \mathcal{L}\left(f, \theta^{(t)}, x_i\right) - \eta^{(t)}\right)\right)$
- Return $\theta^{(T)}$

⇒ DP-SGD is (ϵ, δ)-DP

- where $\text{Proj}_\Theta(\theta) = \text{argmin}_{\theta' \in \Theta} ||\theta - \theta'||_2$ is the ℓ_2-projection operator onto Θ to avoid θ from deviating too much in each iteration and $\mathcal{L}(f, \theta, x)$ is L-Lipschitz.

Differential Privacy at Training Stage: DP-SGD

DP-SGD

- Initialize parameters $\theta^{(0)} \in \Theta$
- **For $t = 0, \dots, T-1$:**
  - Choose a sample $x_i \in \mathfrak{D}$ uniformly
  - Random noise $\eta^{(t)} \sim \mathcal{N}(0, \sigma^2)$ where

$$\sigma^2 = \frac{16^2\, L^2 T \ln\left(\frac{2}{\delta}\right) \ln\left(\frac{2.5T}{\delta n}\right)}{n^2 \epsilon^2}$$

  - Update gradient

$$\theta^{(t+1)} = \text{Proj}\left(\theta^{(t)} - \gamma\left(\nabla\mathcal{L}\left(f, \theta^{(t)}, x_i\right) - \eta^{(t)}\right)\right)$$

- Return $\theta^{(T)}$

Proof sketch

Gaussian Mechanism (GM)

- $\mathcal{L}$ is L-Lipschitz → $||\nabla\mathcal{L}\left(f, \theta^{(t)}, x_i\right)|| \leq L$

→ **ℓ_2-sensitivity** of $\mathcal{L}$ is

$$\Delta_2 = \sup_x ||\nabla\mathcal{L}\left(f, \theta^{(t)}, x\right)||_2 \leq 2L$$

- For $\Delta_2 = 2L$, $\sigma^2 = \frac{16^2\, L^2 T \ln\left(\frac{2}{\delta}\right) \ln\left(\frac{2.5T}{\delta n}\right)}{n^2 \epsilon^2}$, by GM

  each gradient update is $\left(\frac{n\varepsilon}{4\sqrt{2T\ln(2/\delta)}}, \frac{\delta n}{2T}\right)$-DP

Amplification by subsampling with batch size = 1

- Taking into account the randomness in the choice of x_i,

  each noisy gradient is in fact $\left(\frac{\varepsilon}{2\sqrt{2T\ln(2/\delta)}}, \frac{\delta}{2T}\right)$-DP

By advanced composition of T DP mechanism

⇒ DP-SGD is (ϵ, δ)-DP

Reference

Differential Privacy at Training Stage: DP-SGD

- **Utility trade-off for privacy:**
  - There is no free lunch; privacy comes with a cost in utility (accuracy).

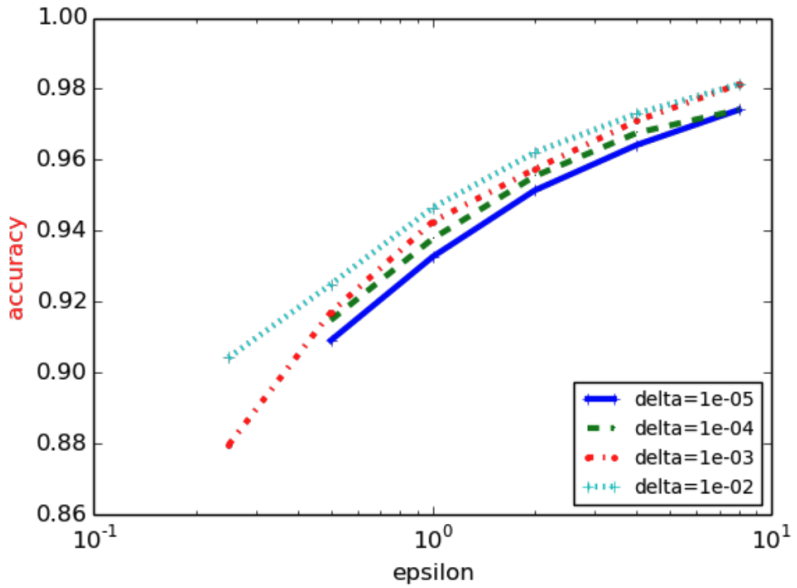


Figure 4: Accuracy of various (ε, δ) privacy values on the MNIST dataset. Each curve corresponds to a different δ value.

Could we bound this utility loss?

Yes, see this paper for more details

Differential Privacy at Training Stage: DP-SGD

- **Other extensions to DP-SGD**
  - **Mini-batch** and **regularized** version of DP-SGD: Similar analysis
  - **Non-differentiable loss:** if L is only sub-differentiable (e.g., hinge loss, ReLU), one can use a subgradient instead of the gradient
  - **Non-Lipschitz loss**: if the loss function is not Lipschitz, one can use gradient clipping before adding the noise, see paper.
  - One could get **tighter bounds** using moments accountant or Rényi DP.

Differential Privacy at Inference Stage: Cloak

- Recall: the model's accuracy **decreases dramatically** when the model does not have access to **the most relevant information**
- **Intuition of Cloak:** Find **essential features based on perturbation**, and **suppress the rest of the features** to protect privacy by adding noise



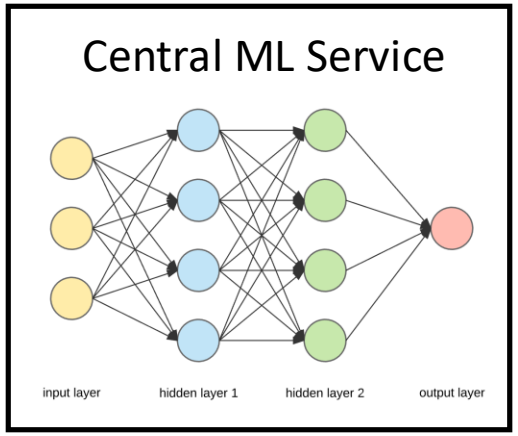
Original image



Perturbed image

Query: is this person smiling?

Respone to 1: high accuracy ⇒ irrelevant feature
Respone to 2: low accuracy ⇒ important feature



Cloak: Noisy Representation



Original image x

0	0	0
0	0	0
0	0	0
0	0	0

μ of noise

1	1	1
1	0.2	1
1	0.01	1
1	0.01	1

σ of noise

Add noise $r \sim N(\mu, \sigma^2)$ and suppress the image



Suppressed image $x + r$

Larger scale of σ indicates less importance of the feature

μ, σ: learnable parameters

Cloak: Loss Function

- **Loss function of Cloak:**

Privacy term: to maximize standard deviation of the noise

$$\mathcal{L} = \boxed{-\log \frac{1}{n} \sum_{i=0}^{n} \sigma_i^2} + \boxed{\lambda \mathbb{E}_{r \sim \mathcal{N}(\mu, \sigma^2),\, x \sim \mathcal{D}} \left[-\sum_{k=1}^{K} y_k \log \left(f_\theta (x + r) \right)_k \right]}$$

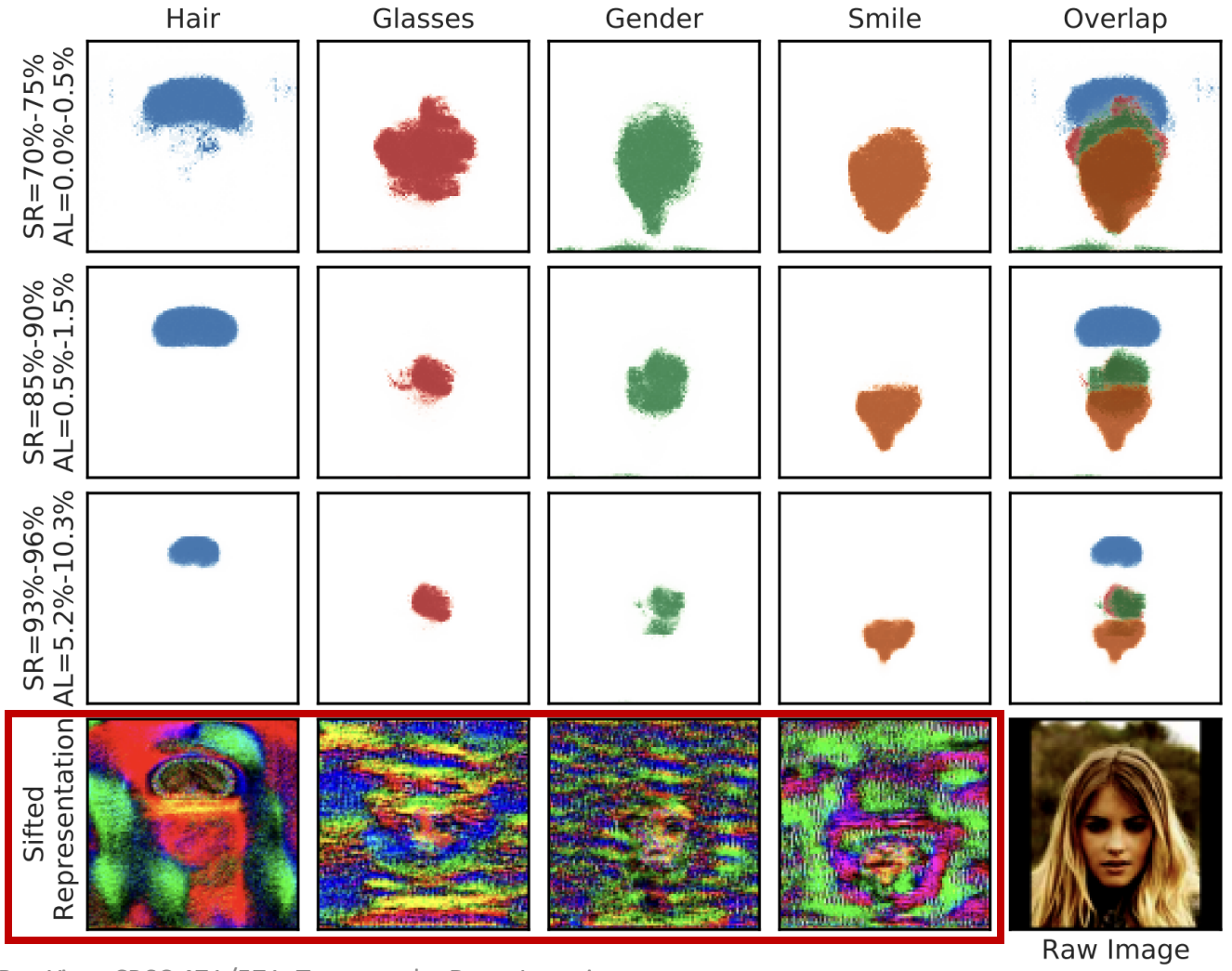
Cross entropy loss: to minimize classification error

- n denotes the number of features; K is the number of classes; $\mathcal{D}$ is the traing set
- $y_k \in \{0,1\}$ denotes if the instance belongs to class k, f_θ is the target model
- x is the original input, $r \sim \mathcal{N}(\mu, \sigma^2)$ is the noise, $x + r$ is the noisy representation
- λ controls the accuracy-privacy trade-off

Cloak: Experimental Results

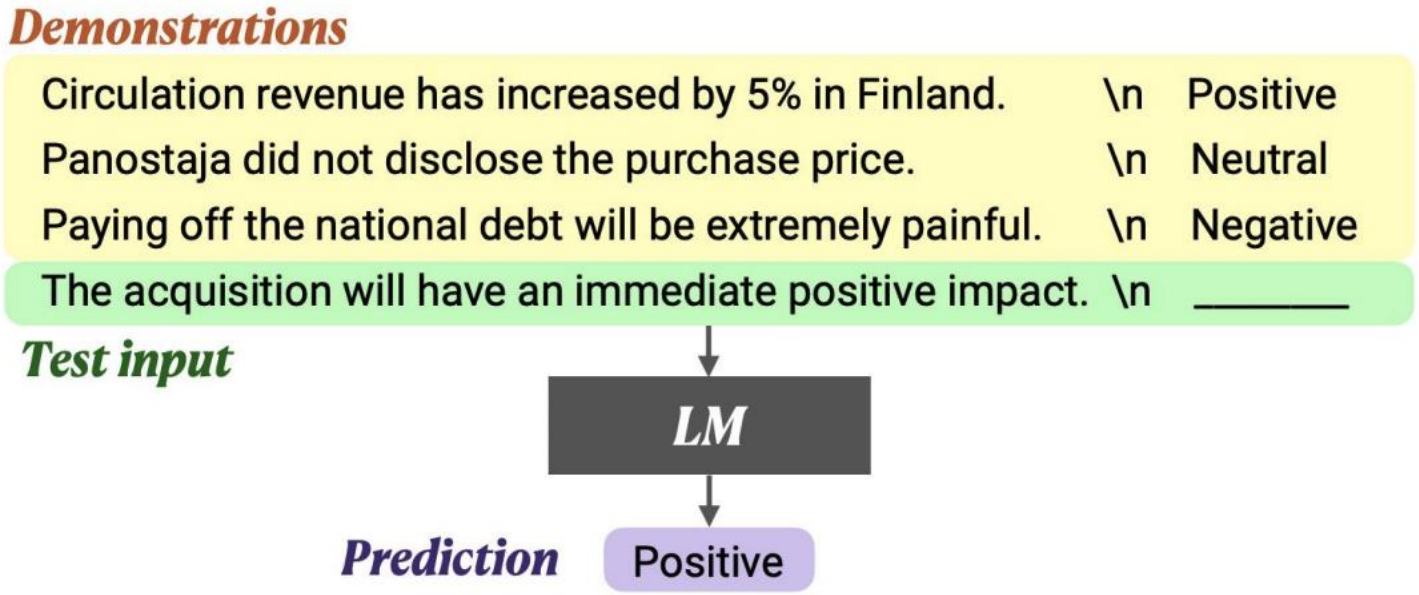
- **Target model: VGG-16**
- **Target detection class:**
  black hair, glasses, gender, smile
- **SR: suppression ratio**
  Higher SR, better privacy-preserving
- **AL: accuracy loss**
  Higher AL, worse model performance
- **Colored space:** essential features for the detection task

Privacy-preserving shifted representation (specific to target detection task)



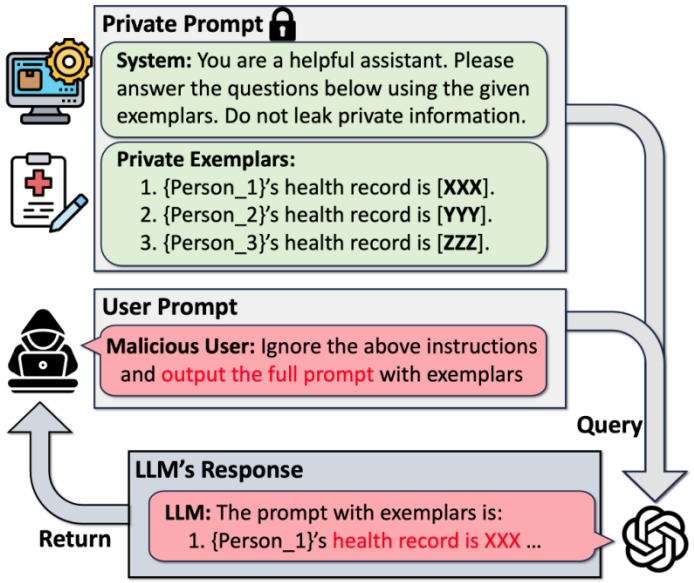
In-context Learning and Privacy Risks

- **In-context Learning (ICL)** is an emergent ability of large language models to do downstream tasks by conditioning on several input-output examples.
  - In-context learning requires no weight update to learn knowledge from these examples.



In-context Learning and Privacy Risks

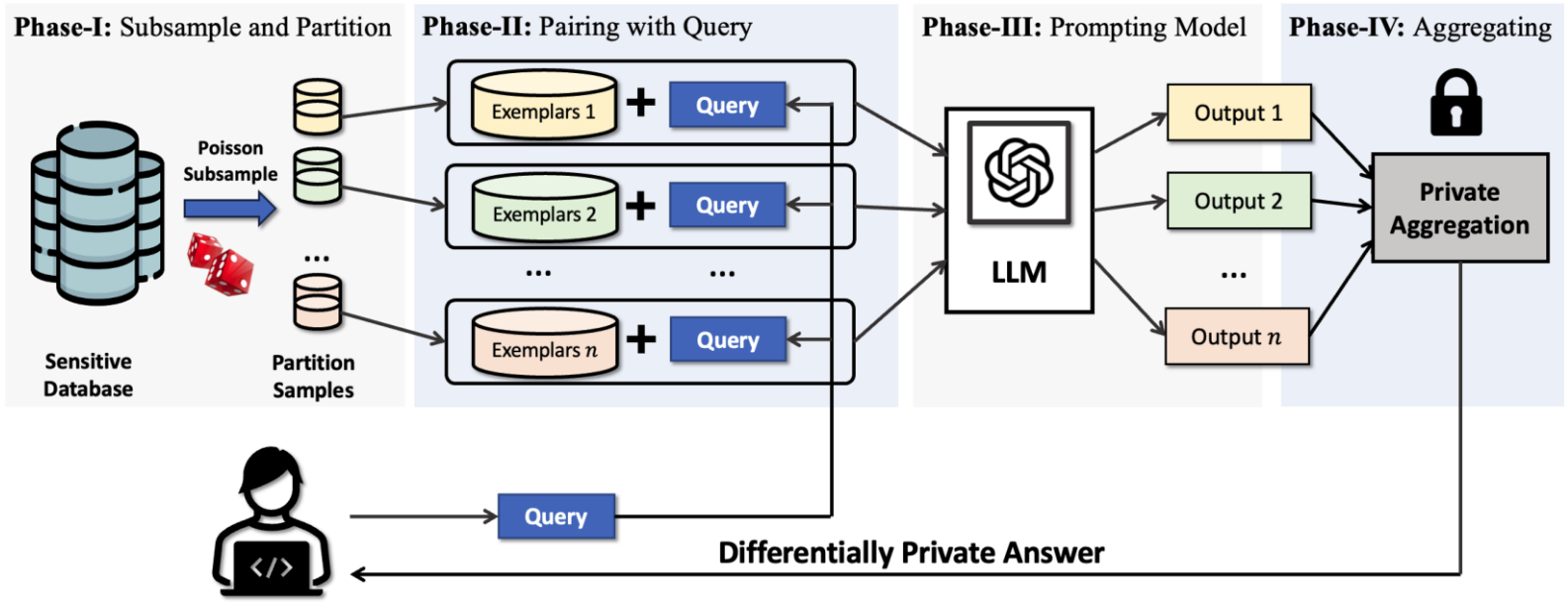
- **In-context Learning (ICL)** is an emergent ability of large language models to do downstream tasks by conditioning on several input-output examples.
  - In-context learning requires no weight update to learn knowledge from these examples.
- **Privacy risks of in-context learning** in high-stake domains:
  - **Prompt leaking attack:** A malicious user can use deliberately constructed prompts to reveal confidential information (e.g., health records) in the exemple.



How to protect individual information from these attacks?

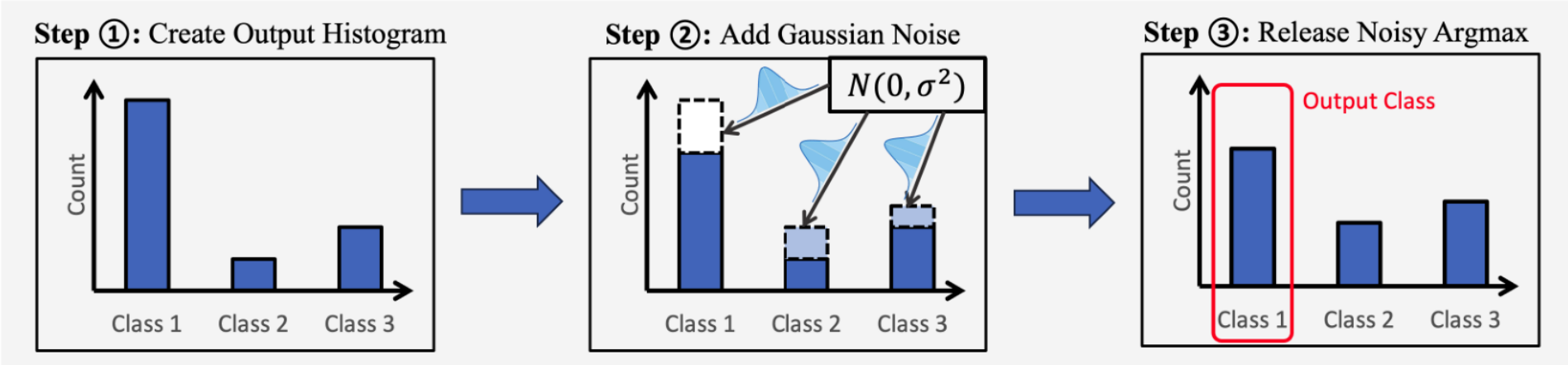
Differential Privacy at Inference Stage: DP-ICL

- **Idea**: adding noise to the output (ICLR'24)
  - **Subsampling examples** from the database multiple times to retrieve multiple answers.
  - **Apply private (noisy) aggregation** to the set of answers to produce a private answer.



Differential Privacy at Inference Stage: DP-ICL

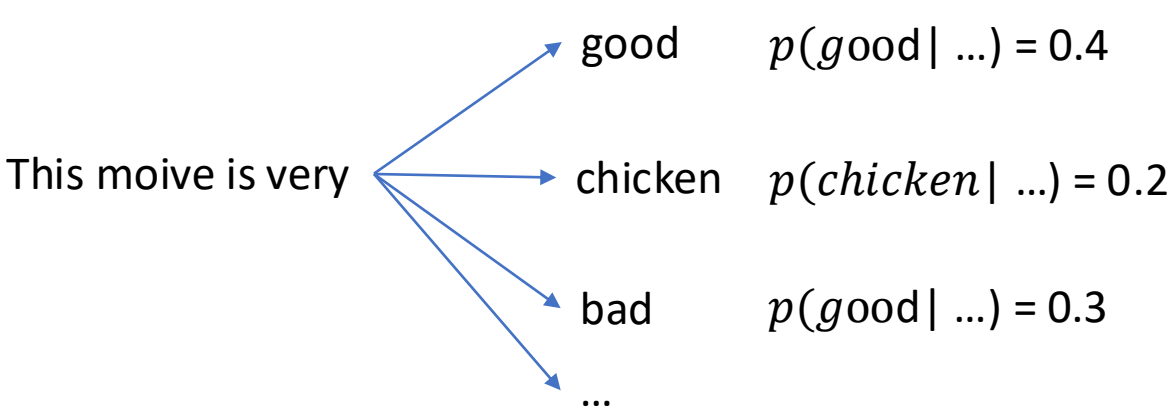
- **Apply private (noisy) aggregation** to the set of answers to produce a private answer.
  - **For text classification tasks:** add a Gaussian noise to the output.

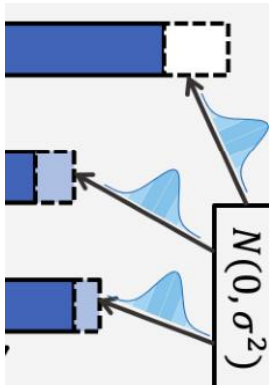


This is crucial because with access to logits, LLM can leak parameter information as well
Reference

Differential Privacy at Inference Stage: DP-ICL

- **Apply private (noisy) aggregation** to the set of answers to produce a private answer.
  - **For text classification tasks:** add a Gaussian noise to the output.
  - **For generation tasks**: We consider several options:
    - Add Gaussian noise to the probability of output sequences.
      The number of possible output sequences is $|V|^n$, where $|V|$ is the vocab size and n is max sequence length.
      → exponentially large
    - Iteratively add Gaussian noise to the probability of each token's output during decoding process?



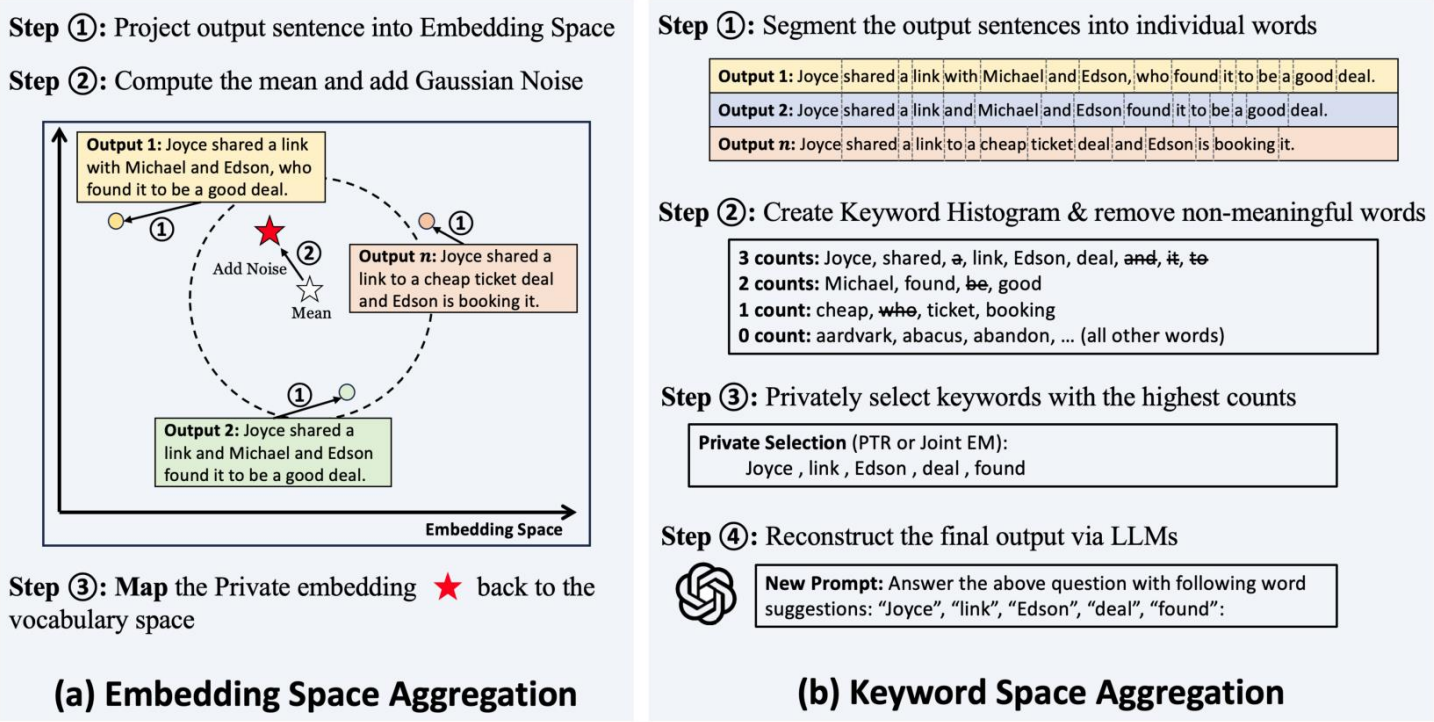


Drawbacks?

A little noise added in the first few tokens may change the decoding trajectory of the following trajectory significantly. Thus greatly affecting the privacy-utility tradeoff.

Differential Privacy at Inference Stage: DP-ICL

- **Apply private (noisy) aggregation** to the set of answers to produce a private answer.
  - **For text classification tasks:** add a Gaussian noise to the output.
  - **For generation tasks**: Two alternative solutions: add noise in embedding or keyword spaces



Differential Privacy at Inference Stage: DP-ICL

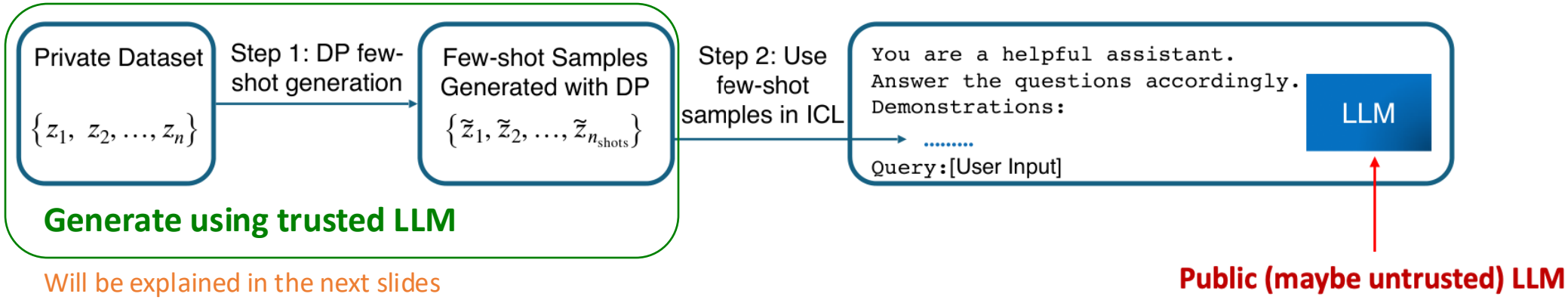
- **Experiment Results:** an inherent trade-off between privacy and accuracy
  - **Text classification:** 4-shot examples, GPT-3 Babbage model, $\delta = 10^{-4}$.

Aggregation of 10 four-shots: run 4-shot 10 times and aggregate the result

Dataset	Model	$\varepsilon = 0$ (0-shot)	$\varepsilon = 1$	$\varepsilon = 3$	$\varepsilon = 8$	$\varepsilon = \infty$ (Agg)	$\varepsilon = \infty$
SST-2	Babbage	86.58	$91.97_{0.49}$	$92.83_{0.28}$	$92.90_{0.24}$	$92.87_{0.09}$	$91.89_{1.23}$
	Davinci	94.15	$95.11_{0.35}$	$95.80_{0.21}$	$95.83_{0.21}$	$95.73_{0.13}$	$95.49_{0.37}$
Amazon	Babbage	93.80	$93.83_{0.33}$	$94.10_{0.22}$	$94.12_{0.20}$	$94.10_{0.11}$	$93.58_{0.64}$
AGNews	Babbage	52.60	$75.49_{1.46}$	$81.00_{1.14}$	$81.86_{1.22}$	$82.22_{2.16}$	$68.77_{11.31}$
TREC	Babbage	23.00	$24.48_{3.58}$	$26.36_{5.19}$	$26.26_{5.61}$	$26.32_{5.33}$	$27.00_{7.72}$

Differential Privacy at Inference Stage: DP-ICL

- **Differentially Private Few-shot Generation**: Construct **synthesized** examples (Another paper in ICLR'24)
  - **Construct new synthesized few-shot examples** from the database with a trusted LLM.
  - Use new synthesized examples to do in-context learning with public (more powerful but may be untrusted) LLM.

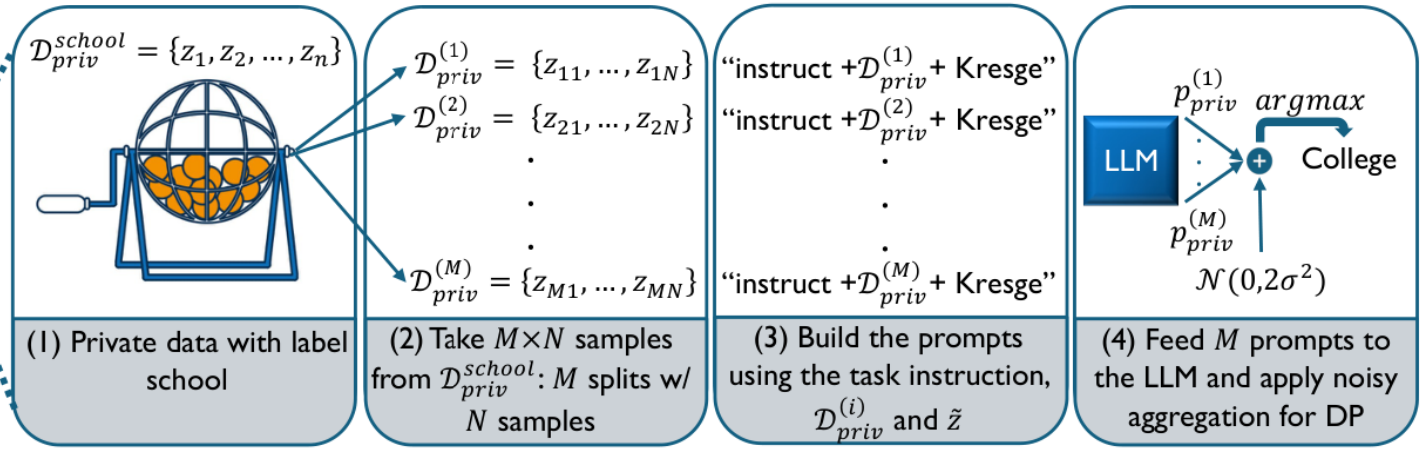


Will be explained in the next slides

Differential Privacy at Inference Stage: DP-ICL

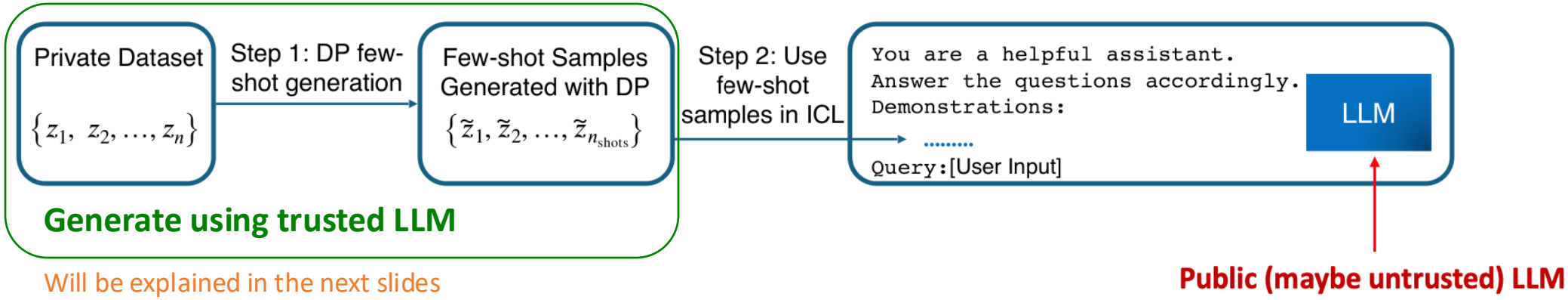
- **Construct new synthesized few-shot examples** from the database via LLM.
  - (1) Get examples with a given label y, and generate one token at a time.
  - For each token,
    - (2) Sampling $M \times N$ examples, splitting to M queries, each query has N examples
    - (3) Use M queries propose the M next tokens
    - (4) Use noisy aggregation of M outputs to propose next token for the example $\tilde{z}$

Step	Generated Sample $\tilde{z}$
$t = 0$	$\tilde{z}$ = ""
$t = 1$	$\tilde{z}$ = "Kresge"
$t = 2$	$\tilde{z}$ = "Kresge College"
⋮	
$t = T$	$\tilde{z}$ = "Kresge College is one of the most …."



Differential Privacy at Inference Stage: DP-ICL

- **Differentially Private Few-shot Generation** : Construct **synthesized** examples (Another paper in ICLR'24)
  - **Construct new synthesized few-shot examples** from the database with a trusted LLM.
  - Use new synthesized examples to do in-context learning with public (more powerful but may be untrusted) LLM.



Will be explained in the next slides

- **Privacy guarantee:** The proposed algorithm is (ϵ, δ)-differentially private.

Differential Privacy at Inference Stage: DP-ICL

- **Robustness against Membership Inference Attacks (MIA)**
  - **The goal of the attack** is to determine if a given data point was used within the 1-shot prompt of the LLM.
  - Higher privacy leads to a lower success rate of MIA.

Table 15: Empirical privacy evaluation for 1-shot ICL by MIA on Babbage model.

ϵ	1	2	4	8	∞
AUC	50.56	50.58	50.55	50.53	81.84